Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- **Introduction to Linear Algebra**
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • 16 min

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Introduction to Linear Algebra

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In this course, we will cover essential mathematical concepts that will provide a solid foundation for future machine learning applications. Numerical data will be represented as vectors, and tables of such data will be represented as matrices—this study is known as linear algebra.

Key Concepts

When we have two vectors representing real-world objects, we need to determine their **similarity**. Similar vectors should yield similar outputs from our machine learning algorithms. To formalize this idea, we use operations that take two vectors as input and return a numerical value representing their similarity, a concept central to analytic geometry.

Some matrix operations are crucial in machine learning as they allow for intuitive data interpretation and more efficient learning. These include fundamental concepts about matrices and matrix decomposition.

Data is often considered a collection of noisy observations of some true underlying signal. Machine learning aims to identify the signal from the noise and provide predictors that express **uncertainty**. This quantification of uncertainty falls under **probability** theory.

To **train** machine learning models, we optimize parameters to maximize performance measures. This process involves optimization to find maxima or minima of functions.

Finally, we use mathematical approaches to revisit the three core components of machine learning: data, models, and parameter estimation. We also provide guidelines for creating experimental setups that avoid overly optimistic evaluations of machine learning systems.

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